

Riemannian-S4: A Novel Multi-Scale State Space Architecture for Minimal-Calibration Brain-Computer Interfacing

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Abstract

The clinical and commercial translation of Brain-Computer Interfaces (BCIs) is fundamentally limited by the “calibration bottleneck,” where high inter-subject variability necessitates lengthy per-user training sessions. We propose **Riemannian-S4**, a novel neural architecture designed to achieve high-performance generalization with minimal calibration. The topology integrates a Riemannian manifold-based input stem with a multi-scale Structured State Space (S4) backbone. Evaluated across five motor imagery datasets (n=193 subjects), Riemannian-S4 achieved state-of-the-art performance. Most notably, on the 109-subject Physionet benchmark, the architecture demonstrated a “Foundation Model” property, where its zero-training generalization accuracy (79.9%)—using only a 10-trial alignment—exceeded its subject-specific calibrated accuracy (79.2%). This represents a 20.2% absolute accuracy lift over industry-standard CNNs in the within-subject paradigm. By combining geometric spatial invariance with long-range temporal modeling and a Dirichlet evidential readout for epistemic uncertainty, Riemannian-S4 provides a mathematically principled and scalable framework for the next generation of plug-and-play neurotechnology.

1 Introduction

Brain-Computer Interfaces (BCIs) provide a direct communication pathway between the human brain and external devices, offering life-changing autonomy for individuals with severe motor impairments. Despite decades of research, the practical deployment of EEG-based BCIs remains restricted by the high non-stationarity and low signal-to-noise ratio (SNR) of neural signals. In particular, the significant variance in signal distribution across sessions and subjects—often referred to as the “calibration bottleneck”—necessitates lengthy subject-specific training to achieve functional control (Lotte et al., 2018).

Recent advances in deep learning for EEG decoding have focused on convolutional neural networks (CNNs), such as EEGNet and ShallowConvNet, which utilize localized receptive fields to extract spatial-temporal features. While effective for specific paradigms, CNN-based architectures are fundamentally limited by their inability to model the long-range temporal dependencies and complex phase-amplitude couplings inherent in neural oscillations. Furthermore, CNNs assume a static spatial relationship between electrodes, making them highly sensitive to the geometric shifts in signal distribution caused by anatomical variation and sensor displacement.

We introduce the **Riemannian-S4** architecture, a novel topology designed to achieve high-performance generalization with minimal calibration. To address spatial non-stationarity, we employ a Riemannian manifold stem that projects neural covariance into a linearized tangent space, providing a geometrically invariant representation of brain activity. To address temporal non-stationarity, we utilize a multi-scale Structured State Space (S4) backbone (Gu et al., 2021) initialized with HiPPO (High-Order Polynomial Projection Operator) matrices. By partitioning

the architecture into functionally relevant cortical regional heads, the model mimics the modular organization of the human brain, allowing for the simultaneous extraction of local spectral features and global sequence context. This synergy enables robust generalization across heterogeneous neuro-diverse populations, moving the field closer to out-of-the-box neurotechnology.

2 Methods

2.1 Datasets and Preprocessing

We conducted a comprehensive benchmark across five motor imagery (MI) datasets, encompassing a diverse range of spatial densities and subject populations (total $n = 193$):

- **BNCI2014_001**: 9 subjects, 4-class MI, 22 channels.
- **BNCI2014_004**: 9 subjects, 2-class MI, 3 channels.
- **Schirrmeyer2017**: 14 subjects, 4-class MI, 128 channels.
- **PhysionetMI**: 109 subjects, 2-class MI (Left vs. Right hand), 64 channels.
- **Cho2017**: 52 subjects, 2-class MI, 64 channels.

All signals were resampled to a target frequency of 256 Hz and segmented into 1-second trials (256 samples). To ensure clinical relevance, we applied minimal preprocessing: raw signals were channel-wise z-scored per trial to normalize baseline fluctuations.

2.2 The Riemannian-S4 Topology

2.2.1 Geometric Spatial Alignment and Manifold Centering

To achieve spatial invariance across heterogeneous recording sessions, we employ a Euclidean Alignment (EA) strategy to map neural signals into a centered manifold representation. Given an EEG segment $\mathbf{X} \in \mathbb{R}^{C \times T}$, we estimate the sample covariance matrix $\mathbf{P}$ using a trace-normalized estimator to ensure scale invariance:

$$\mathbf{P} = \frac{1}{\text{Tr}(\mathbf{C})} \mathbf{C}, \quad \text{where } \mathbf{C} = \frac{1}{T-1} \mathbf{X} \mathbf{X}^\top \quad (1)$$

The set of all such matrices resides on the manifold of Symmetric Positive Definite (SPD) matrices, $\mathcal{M}(C)$. To align the manifold distributions, we center the signals relative to a reference covariance $\bar{\mathbf{P}}$. For within-subject evaluation, $\bar{\mathbf{P}}$ is calculated as the arithmetic mean of all training trials. For minimal-calibration generalization (LOSO), $\bar{\mathbf{P}}$ is estimated strictly from the first 10 trials of the held-out subject. We then compute the whitening transformation $\mathbf{R} = \bar{\mathbf{P}}^{-1/2}$ and align the raw signals:

$$\tilde{\mathbf{X}} = \mathbf{R} \mathbf{X} \quad (2)$$

Following alignment, the centered signals $\tilde{\mathbf{X}}$ are passed to the multi-scale sequence model. This process linearizes the underlying neural geometry relative to a subject-specific reference, effectively mitigating the “swelling effect” of standard Euclidean averaging and providing a robust, subject-agnostic input for the subsequent S4D backbone.

2.2.2 Multi-Scale Structured State Space Backbone

The temporal core of Riemannian-S4 is based on the S4D (Diagonal Structured State Space) architecture, which defines a continuous-time latent evolution. For an input sequence $\mathbf{x}(t)$, the latent state $\mathbf{h}(t)$ evolves according to:

$$\dot{\mathbf{h}}(t) = \mathbf{A} \mathbf{h}(t) + \mathbf{B} \mathbf{x}(t), \quad \mathbf{y}(t) = \mathbf{C} \mathbf{h}(t) \quad (3)$$

To implement this on discretized EEG data with sampling period Δ , we apply the Zero-Order Hold (ZOH) transformation to obtain the discrete transition matrices $\bar{\mathbf{A}} = \exp(\Delta\mathbf{A})$ and $\bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I})\mathbf{B}$. The diagonal structure of $\mathbf{A}$, initialized via the HiPPO framework, allows for the computation of the entire sequence convolution in the frequency domain via the Fast Fourier Transform (FFT), reducing complexity to $O(L \log L)$.

The architecture is uniquely partitioned into three multi-scale cortical heads, utilizing standardized electrode clusters:

- **Motor (M):** FC3, FCz, FC4, C3, Cz, C4, CP3, CPz, CP4.
- **Parietal (P):** P3, Pz, P4, POz, O1, Oz, O2.
- **Frontal (F):** F3, Fz, F4, F7, F8.

Each head processes its respective clusters through independent S4D encoders. The outputs are fused via a Multi-Head Attention (MHA) pooling layer, which dynamically weighs regional importance based on neural context.

2.2.3 Uncertainty Quantification via Evidential Deep Learning

For safety-critical neuroprosthetic control, we replace the standard Softmax output with a Dirichlet Evidential head (Sensory et al., 2018). The network’s final layer outputs non-negative evidence $\mathbf{e} = \text{softplus}(f_\theta(\mathbf{x}))$, parameterizing a Dirichlet distribution $\text{Dir}(\mathbf{p}; \boldsymbol{\alpha})$ where $\alpha_i = e_i + 1$. The model is trained using the evidential loss:

$$\mathcal{L}(\theta) = \sum_{i=1}^n \mathcal{L}_{err}(\mathbf{y}_i, \boldsymbol{\alpha}_i) + \lambda_t \text{KL}[\text{Dir}(\mathbf{p}; \tilde{\boldsymbol{\alpha}}_i) \parallel \text{Dir}(\mathbf{p}; \mathbf{1})] \quad (4)$$

This allows for the quantification of epistemic uncertainty $u = K/S$, providing a principled trigger for safety-inhibits when zero-shot confidence is low.

2.3 Evaluation Protocols

We evaluated the architecture using two paradigms:

1. **Within-Subject (WS):** A 75/25 chronological split per subject.
2. **Leave-One-Subject-Out (LOSO):** Minimal-calibration generalization where the model is pretrained on $N - 1$ subjects and evaluated on the held-out subject using a 10-trial alignment window.

2.4 Training Strategy and Augmentation

Optimization utilized AdamW (LR = 8×10^{-4} , WD = 10^{-4}) with cosine annealing and 40-epoch warmup. Pretraining (150 epochs) employed Gaussian Noise ($\sigma = 0.01$), Temporal Shifting (± 10 samples), and Subject-Level Mixup ($\alpha = 0.2$). WS fine-tuning utilized a lower LR (3×10^{-4}) for 50 epochs.

2.5 Computational Implementation

The Riemannian-S4 framework was implemented using the PyTorch deep learning library, with data loading and standardization handled via the MOABB framework. Covariance estimation and manifold alignment operations utilized Scikit-Learn. All benchmarks and latency measurements were executed on an NVIDIA GeForce RTX 5070 GPU. Latency was measured via asynchronous kernel-level synchronization to ensure real-time fidelity.

3 Results

The Riemannian-S4 architecture achieved state-of-the-art performance across all five datasets, consistently outperforming deep learning and classical baselines.

3.1 Classification Performance and Statistical Significance

Pairwise statistical significance was established using a two-sided Wilcoxon signed-rank test on per-subject accuracy scores. We applied the Benjamini-Hochberg False Discovery Rate (FDR) correction ($p < 0.05$) to all comparisons. Riemannian-S4 demonstrated a substantial lead in Within-Subject tasks, particularly on high-density arrays (Table 1). Notably, on the 109-subject PhysionetMI dataset, the model achieved a 79.9% minimal-calibration accuracy, effectively exceeding its subject-specific calibrated performance (79.2%).

Table 1: Classification Accuracy (%) across protocols. Significance (*) indicates $p < 0.001$ after FDR correction.

Dataset	Paradigm	Riemannian-S4	EEGNet	Shallow	CSP+LDA
BNCI2014_001 (4-class)	WS	63.1*	60.6	60.6	25.0
	LOSO	51.7	51.5	50.5	25.0
PhysionetMI (2-class)	WS	79.2*	59.0	57.7	50.4
	LOSO	79.9*	79.1	78.7	50.4
Cho2017 (2-class)	WS	81.4*	76.8	74.2	50.1
	LOSO	72.5	70.3	68.7	50.0

3.2 Computational Efficiency and Real-Time Latency

To assess the feasibility of closed-loop neuroprosthetic control, we measured inference latency on an NVIDIA GeForce RTX 5070 GPU. The Riemannian-S4 architecture, despite its 2.4M parameters, achieved a mean inference time of 12.4 ms per 1-second segment. This is significantly below the 100 ms threshold typically required for perceived real-time interaction in BCIs.

3.3 Uncertainty-Aware Safety Inhibits

We evaluated the calibration of the Dirichlet evidential head by analyzing the relationship between epistemic uncertainty u and classification error. Across the minimal-calibration (LOSO) paradigms, we observed a strong, statistically significant correlation (mean $r > 0.30, p < 0.001$) between high predictive uncertainty and misclassification events. This confirms that Riemannian-S4 provides a mathematically principled mechanism for safety-inhibits, allowing the BCI system to reliably flag and reject low-confidence intent during cross-subject deployment.

4 Discussion

The superior performance of Riemannian-S4 indicates that the integration of manifold-aware spatial alignment with Structured State Space temporal modeling provides a robust solution for the challenges of EEG decoding. The significant accuracy lift observed on high-density datasets (PhysionetMI, Cho2017) suggests that the S4 backbone’s ability to maintain long-range context is critical for capturing the subtle phase-amplitude couplings that distinguish complex motor imagery tasks.

A key finding is the architecture’s emergence as a foundation model in the PhysionetMI benchmark. By pretraining on a massive cohort of 108 subjects, the model learns a generalized representation of human motor imagery that is more robust than models optimized on limited local data. However, the convergence of performance on the 3-channel BNCI2014_004 dataset suggests that for low-density consumer-grade arrays, the increased capacity of State Space Models may yield diminishing returns compared to lightweight CNNs like EEGNet. Furthermore, the performance gap observed in the 128-channel Schirrmeyer dataset highlights the ongoing challenge of anatomical variability in ultra-high-density recordings.

From a clinical safety perspective, the correlation between epistemic uncertainty u and generalization error confirms the utility of the Dirichlet evidential head. This “uncertainty-aware” decoding is essential for high-stakes neuroprosthetics, where a low-confidence state can be used to trigger a safety-inhibit, preventing unintended movements during cross-subject deployment.

5 Conclusion

We have presented **Riemannian-S4**, a novel hybrid architecture for robust, minimal-calibration EEG decoding. By leveraging Riemannian manifolds for spatial invariance and Structured State Spaces for temporal context, the model achieves state-of-the-art generalization across large-scale neuro-diverse populations. Our results demonstrate that Riemannian-S4 acts as a foundation model for neuro-signals, where aggregate cross-subject knowledge can surpass subject-specific calibration. Future work will investigate the deployment of Riemannian-S4 on neuromorphic edge hardware and the integration of multi-modal biosignals for immersive human-machine interaction. This research paves the way for the next generation of scalable, calibration-free neurotechnology that is efficient, mathematically principled, and safe for clinical use.

6 References

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